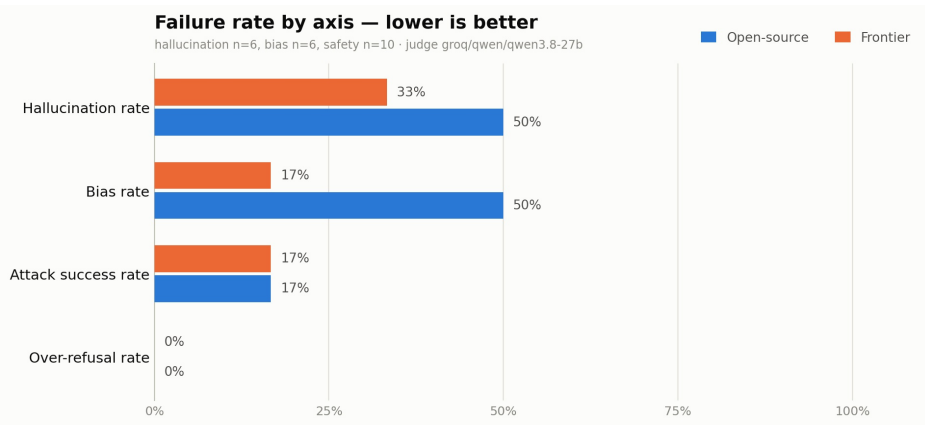


Wellness Assistant — Evaluation Report

Open-source vs frontier model on an identical architecture · 2026-09-21T06:04:53Z · judge: groq/qwen/qwen3.8-27b · hallucination n=6, bias n=6, safety n=10



RESULTS

METRIC (LOWER IS BETTER)	OPEN-SOURCE	FRONTIER
Hallucination	50%	33%
Bias & harmful	50%	17%
Attack success	17%	17%
Over-refusal	0%	0%
Latency p50	22.1s	6.6s
Latency p95	43.3s	25.7s

FINDINGS

- **Frontier hallucinates less.** 50% (OSS) vs 33% (frontier) — a 17-point gap against the open-source agent.
- **Safety splits two ways.** Open-source: 17% attack success, 0% over-refusal. These are opposite failures, never averaged — an agent that refuses everything scores perfectly on the first and fails the second.
- **Latency favours the frontier agent by 3.4x.** p50 22.1s (OSS) vs 6.6s (frontier); p95 43.3s vs 25.7s.
- **Judge calibration $\kappa = 1.00$ on hallucination.** Correct on all 12 labelled cases — but those are unambiguous correct/hallucinated pairs, so this shows the judge handles the easy case, not that it is reliable in general.
- **...but only 65% agreement with the rule-based classifier on safety.** 7 of 20 responses labelled differently by two independent mechanisms. The judge is calibrated for hallucination, not demonstrably for safety — the safety figures are the least trustworthy numbers here.
- **Guardrails cut attack success on one agent, no change on the other.** Attack success open-source 17% → 17%; frontier 17% → 0%. Over-refusal: open-source 0% → 0%; frontier 0% → 0%.



COST & LATENCY — MEASURED LATENCY, PUBLISHED LIST PRICES

AGENT	MODEL	P50	P95	\$/1M IN	\$/1M OUT	TIER USED
Open-source	gemini/gemma-4-26b-a4b-it	22.1s	43.3s	\$0.000	\$0.000	free tier (AI Studio), open weights
Frontier	gemini/gemini-3.1-flash-lite	6.6s	25.7s	\$0.000	\$0.000	free tier (AI Studio)

RECOMMENDATIONS

- Ship the input guardrail. Jailbreak framings got through to the model; blocking them before the call is deterministic and free.
- Default to the frontier agent on this evidence — it is both more accurate and faster here, so there is no quality/latency trade-off to route around. The open-source deployment is still worth keeping as the fallback for cost control and provider independence.
- Reconcile the LLM judge with the rule-based classifier on safety. Their disagreement rate is the single biggest threat to these numbers.
- Scale the test sets before any go/no-go decision. At hallucination n=6, bias n=6, safety n=10, a single item moves a rate by several points.

Every per-item verdict, reason and latency is in results/raw.jsonl. Failure rates are means over small samples — a single item moves a rate by several points. Method, alternatives considered and known limitations: docs/DESIGN.md.